Day 6 Task - Trace a Request

Tekloomy - Cyber Security Internship - Module 2 - Day 6 | Tap a box and type. Pre-filled boxes can be edited.

Name: AKASHA DINESHA HOSURA **Date:** 6/11/2026

Step 1 - Who am I?

Windows: ipconfig /all

Your private IPv4 address	10.155.196.152
Subnet mask	255.255.255.0
Default gateway (router) IP	10.155.196.204
DNS server(s) your device uses	10.155.196.204
MAC address of Wi-Fi adapter (first 3 pairs only, e.g. 3C:5A:B4)	E4-C7-67
Manufacturer of adapter (macvendors.com)	

Step 2 - Is the network alive? (ping)

Record average round-trip time in ms, or 'no reply'

ping <your gateway IP> - Can I reach my router?	5ms
ping 8.8.8.8 - Can I reach the internet by IP?	56ms
ping google.com - Does DNS + internet work?	42ms

2D. If ping 8.8.8.8 works but ping google.com fails, what is broken?

That DNS is working because google.com successfully resolved to an ipv6 address and replied.

Step 3 and Step 4

Windows: tracert google.com | nslookup <name>

Step 3 - Which road does it take? (traceroute)

How many hops to reach google.com?	8 hops
IP of hop 1 (should match your gateway)	2401:4900:19d7:265::ef
First hop that is NOT a private IP (ISP edge)	Hop 1 -2401:4900-19d:265::ef
Biggest time jump: which hop? Why might that be?	Hop no.: 8 Time: 217 ms. A big jump usually means a long physical distance (e.g. ISP to an intern
How many hops showed * * *? Why do routers do that?	1 hop(Hop 5)

Step 4 - Ask the phone book (nslookup)

google.com	IP address(es) returned 2404:6800:4007:830:200e	DNS server that answered 10:155:196:204
tekloomy.com	IP address(es) returned 216.198.79.1	DNS server that answered 10.155.196.204
College/employer site:	IP address(es) returned 2606:4700:3033::6815:adf	DNS server that answered 10.155.196.204

4D. nslookup google.com 1.1.1.1 - same IPs? Why might two DNS servers answer differently?

Same IPs? (Yes / No): NO. Two DNS servers can return different ip address becose google user multipl

Step 5 - Who is my laptop talking to? (netstat)

Windows: netstat -ano (open a website first)

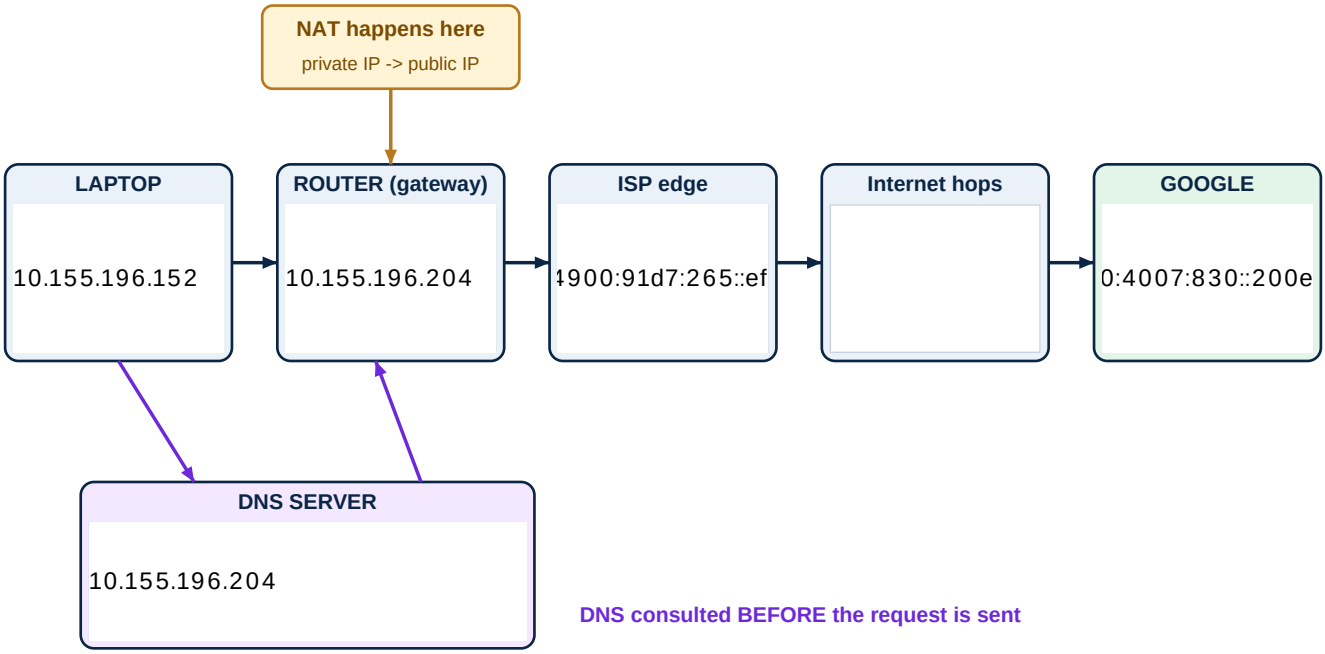
How many connections are ESTABLISHED?	29
Pick one remote IP:port (e.g. 142.250.x.x:443). What service is port 443?	104.199.241.202:443
Any LISTENING ports on your machine? List up to 3.	135,445,3306
Look up one remote IP at ipinfo.io. Which company owns it?	Google

Step 6 and Step 7

Draw the path, then answer the troubleshooting question

Step 6 - Draw the path

Type your IPs in the boxes. NAT and DNS are marked for you.



Step 7 - 'The website is down': what do you check?

.....